

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

QUADRATIC EQUATIONS

DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)



DIBY-01

If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ in terms of a, b, c is

(A) $\frac{b^2-2ac}{ac}$

(B) $\frac{2ac-b^2}{ac}$

(C) $\frac{b^2-2ac}{a^2c^2}$

(D) $\frac{b^2}{2}$

[Ans.: c]

DIBY-02

Let α and β be the roots of the quadratic equation $(x-2)(x-3) + (x-3)(x+1) + (x+1)(x-2) = 0$. Find the value of

$$\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)} + \frac{1}{(\alpha-3)(\beta-3)}$$

[Ans.: 0]

DIBY-03

If α, β are the roots of $ax^2 + bx + c = 0, (a \neq 0)$ and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0, (A \neq 0)$ for some constant δ , then prove that, $\frac{b^2-4ac}{a^2} = \frac{B^2-4AC}{A^2}$. [JEE-2000]

DIBY-04

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of

$$\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}} \text{ is}$$

[Ans.: D]

(A) 1

(B) 3

(C) 4

(D) 2

DIBY-05

Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. The quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}(\alpha^6 + \beta^6)$, is:

[JEE Mains-2024]

(A) $x^2 - 190x + 9466 = 0$

(B) $x^2 - 195x + 9466 = 0$

(C) $x^2 - 195x + 9506 = 0$

(D) $x^2 - 180x + 9506 = 0$

[Ans.: C]

DIBY-06

If the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ has equal roots, where $a + c = 15$ and $b = \frac{36}{5}$, then $a^2 + c^2$ is equal to

[JEE Mains-2025]

[Ans. : 117]

DIBY-07

Let $\alpha \in \mathbb{R}$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is

[JEE Mains-2023]

[Ans. : 45]

DIBY-08

If 2 and 6 are the roots of the equation $ax^2 + bx + 1 = 0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is :

[JEE Mains-2024]

(A) $x^2 + 8x + 12 = 0$

(B) $2x^2 + 11x + 12 = 0$

(C) $4x^2 + 14x + 12 = 0$

(D) $x^2 + 10x + 16 = 0$

[Ans. : A]

NEWTON'S FORMULA

DIBY-09

Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n, n = 1, 2, 3 \dots$, then
[JEE mains-2020]

(A) $6S_6 + 5S_5 = 2S_4$

(B) $6S_6 + 5S_5 + 2S_4 = 0$

(C) $5S_6 + 6S_5 = 2S_4$

(D) $6S_6 + 6S_5 + 2S_4 = 0$

[Ans. : C]

DIBY-10

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then
[JEE Mains-2024]

(A) $2S_{12} = S_{11} + S_{10}$

(B) $S_{12} = S_{11} + S_{10}$

(C) $2S_{11} = S_{12} + S_{10}$

(D) $S_{11} = S_{10} + S_{12}$

[Ans. : B]

DIBY-11

Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to

[JEE Mains-2024]

[Ans. : 4]

NEWTON'S FORMULA

DIBY-12

Let $\alpha, \beta; \alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n, n \in N$. Then $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ is equal to: **[JEE Mains-2024]**

- (A) $10\sqrt{2}P_9$ (B) $10\sqrt{3}P_9$ (C) $11\sqrt{2}P_9$ (D) $11\sqrt{3}P_9$ **[Ans.: B]**

DIBY-13

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is: **[JEE Mains-2021]**

- (A) 2 (B) 1 (C) 4 (D) 3 **[Ans.: A]**

DIBY-14

Let $\alpha, \beta (\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to. **[JEE Mains-2022]**

[Ans.: 16]

QUADRATIC INVOLVING MODULUS

REVISE: BASIC MATHEMATICS



DIBY-15

The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is:

[JEE Mains-2021]

- (A) 1 (B) 4 (C) 3 (D) 2

[Ans.: D]

DIBY-16

The sum, of the squares of all the roots of the equation $x^2 + |2x - 3| - 4 = 0$, is

[JEE Mains-2025]

- (A) $6(2 - \sqrt{2})$ (B) $3(3 - \sqrt{2})$ (C) $6(3 - \sqrt{2})$ (D) $3(2 - \sqrt{2})$

[Ans.: A]

DIBY-17

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is:

[JEE Mains-2020]

- (A) $\frac{25}{9}$ (B) $\frac{25}{81}$ (C) $\frac{5}{9}$ (d) $\frac{5}{27}$

[Ans.: B]

QUADRATIC INVOLVING MODULUS

REVISE: BASIC MATHEMATICS



DIBY-18

The sum of the squares of the roots of $|x + 2|^2 + |x - 2| - 2 = 0$ and the squares of the roots of $x^2 - 2|x - 3| - 5 = 0$, is **[JEE Mains-2025]**

- (A) 26 (B) 36 (C) 30 (d) 24 **[Ans.: B]**

DIBY-19

The number of real roots of the equation $x|x| - 5|x + 2| + 6 = 0$ **[JEE Mains-2023]**

- (A) 5 (B) 3 (C) 6 (D) 4 **[Ans.: B]**

DIBY-20

The sum of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is: **[JEE Mains-2023]**

- (A) $9 + \sqrt{3}$ (B) $11 + \sqrt{3}$ (C) $9 - \sqrt{3}$ (D) $11 - \sqrt{3}$ **[Ans.: A]**

DIBY-21

The number of real solutions of the equation $x|x + 5| + 2|x + 7| - 2 = 0$ is **[JEE Mains-2024]**

[Ans.: 3]

DIBY-22

Let $p, q \in \{1, 2, 3, 4\}$. Find the number of equations of the form $px^2 + qx + 1 = 0$ having real roots.

[Ans. : 7]

DIBY-23

If α, β are the roots of equation $x^2 + x - 2 = 0$, then find the value of $\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2 + \beta^2 + \alpha + \beta}$.

DIBY-24

Find the range of $f(x)$:

(i) $f(x) = 2x^2 - 3x + 2$

Ans. : $\left[\frac{7}{8}, \infty\right)$

(ii) $f(x) = 2x^2 - 3x + 2, x \in [0, 2]$

Ans. : $\left[\frac{7}{8}, 4\right]$

(iii) $f(\theta) = 2\cos^2\theta - 6\sin\theta + 1$

Ans. : $[-5, 7]$

DIBY-25

Find range of following functions:

(i) $f(x) = 3x^2 - 2x - 7$

Ans: $[-22/3, \infty)$

(ii) $f(x) = 3x^2 - 2x - 7, x \in (0, 5]$

Ans: $[-22/3, 58]$

(iii) $f(x) = 3x^2 - 2x - 7, x \in [-6, -1]$

Ans: $[-2, 113]$

DIBY-26

Show that if x is real then $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$ lies from $1/7$ and 7 .

DIBY-27

We call ' p ' a good number if the inequality $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq p$ is satisfied for any real x . Find the smallest integral good number.

[Ans. : 4]

DIBY-28

Let the sum of the maximum and the minimum values of the function

$f(x) = \frac{2x^2 - 3x + 8}{2x^2 + 3x + 8}$ be $\frac{m}{n}$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to **[JEE Mains-2024]**

- (A) 182 (B) 217 (C) 195 (D) 201

Ans.: (D)

DIBY-29

The number of real solutions of the equation $3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$, is

[JEE Mains-2023]

- (A) 4 (B) 0 (C) 3 (D) 2

Ans.: (B)

DIBY-30

If $x^2 + 4ax + 3 = 0$ and $2x^2 + 3ax - 9 = 0$ have a common root then the value of $|a|$ is

- (A) 1 (B) 0 (C) -1 (D) None of these

[Ans.: A]

DIBY-31

If a, b, p, q are non-zero real numbers, then the equations $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have:

- (A) No common root
- (B) One common root if $2a^2 + b^2 = p^2 + q^2$
- (C) Two common roots if $3pq = 2ab$
- (D) Two common roots if $3qb = 2ap$

[Ans. : A]

DIBY-32

$x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

- (A) 24
- (B) 0, 24
- (C) 3, 24
- (D) 0, 3

[Ans. : B]

DIBY-33

If $(x^2 + px + 1)$ is a factor of $(ax^3 + bx + c)$, then

- (A) $a^2 + c^2 = -ab$
- (B) $a^2 - c^2 = -ab$
- (C) $a^2 - c^2 = ab$
- (D) None of these

[Ans. : C]

DIBY-34

Let $\lambda \neq 0$ be in R . If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to: **[JEE MAINS - 2020]**

(A) 18

(B) 36

(C) 9

(D) 27

Ans. A

DIBY-35

If one root is square of the other root of equation $x^2 + px + q = 0$, then the relation between p and q is **[IIT-JEE-2004]**

(A) $p^3 - q(3p - 1) + q^2 = 0$

(B) $p^3 - q(3p + 1) + q^2 = 0$

(C) $p^3 + q(3p - 1) + q^2 = 0$

(D) $p^3 + q(3p + 1) + q^2 = 0$

Ans. A

DIBY-36

If the quadratic equation $3x^2 + ax + 1 = 0$ & $2x^2 + bx + 1 = 0$ have a common root find the value of $5ab - 2a^2 - 3b^2$, $(a, b, \in R)$.

Ans. 01

DIBY-37

Let $x^2 - (m - 3)x + m = 0 (m \in R)$ be a quadratic equation. Find the value of m for which

- (i) both the roots are smaller than 2.
- (ii) both the roots are greater than 2.
- (iii) one root is smaller than 2 and the other root is greater than 2.
- (iv) exactly one root lies in the interval $(1, 2)$.
- (v) both the roots lie in the interval $(1, 2)$.
- (vi) one root is greater than 2 and the other root is smaller than 1.
- (vii) atleast one root lie in interval $(1, 2)$.
- (viii) atleast one root is greater than 2.

Ans.: (i) $m \in (-\infty, 1]$, (ii) $m \in [9, 10)$, (iii) $m \in (0, \infty)$, (iv) $m \in (10, \infty)$, (v) $m \in \phi$, (vi) $m \in \phi$
 (vii) $m \in (10, \infty)$, (viii) $m \in [9, \infty) - \{10\}$

DIBY-38

Find all values of a for which the equation $4x^2 - 2x + a = 0$ has two roots lie in the interval $(-1, 1)$.

$$\text{Ans. : } a \in \left(-2, \frac{1}{4}\right]$$

DIBY-39

The number of integral values of k , for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is **[JEE Mains-2023]**

- (A) 2 (B) 0 (C) 1 (D) 3

Ans. : (C)

DIBY-40

If α, β, γ are roots of $5x^3 - qx - 1 = 0, (q \in R)$ find the value of $\frac{\alpha^2-3}{\beta\gamma} + \frac{\beta^2-3}{\alpha\gamma} + \frac{\gamma^2-3}{\alpha\beta}$.

Ans. : 03

DIBY-41

Given that α, β, γ are the roots of cubic equation $x^3 - 3x^2 + 2x + \frac{2}{3} = 0$.

The value of $\alpha^4 + \beta^4 + \gamma^4$ is equal to

Ans. : 09

DIBY-42

$P(x) = x^3 + 33x^2 + 327x + 935$. Let $P(x)$ be a polynomial as described above with $a; b; c$ the roots of $P(x)$. Find $a^2 + b^2 + c^2$ without solving $P(x) = 0$.

If one root is square of the other root of equation $x^2 + px + q = 0$, then the relation between p and q is **[IIT-JEE-2004]**

[Ans. : A]

A $p^3 - q(3p - 1) + q^2 = 0$

B $p^3 - q(3p + 1) + q^2 = 0$

C $p^3 + q(3p - 1) + q^2 = 0$

D $p^3 + q(3p + 1) + q^2 = 0$

DIBY-44

If α and β are the roots of the quadratic equation $x^2 + x\sin\theta - 2\sin\theta = 0$,

$\theta \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12})(\alpha - \beta)^{24}}$.

[JEE Mains-2019]

[Ans.: c]

A $\frac{2^6}{(\sin\theta + 8)^{12}}$

B $\frac{2^{12}}{(\sin\theta - 4)^{12}}$

C $\frac{2^{12}}{(\sin\theta + 8)^{12}}$

D $\frac{2^{12}}{(\sin\theta - 8)^6}$



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**